Evaluating Sample-Based Quantum Diagonalization

Molecular Ground-State Energy Estimation: Sampling, Subspace Selection, Repeatability and Simulated Noise

Qiskit Fall Fest 2026 · Full SQD Pipeline Challenge

Team CG

Glory Ogunsola | Chiagoziem Mushinwa

Implementation: Google Colab
Molecular systems: water (H_2O) and lithium hydride (LiH)

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Abstract

This project investigates a Sample-Based Quantum Diagonalization (SQD) procedure to estimate ground state energies of small molecular systems. Using the STO-3G basis set and a Qiskit-based simulation workflow in Google Colab, the study considers shot-count and subspace settings, repeatability across random seeds, and the effect of synthetic bit-flip perturbations on sampled configurations. The report presents five-seed results for the H_2O and LiH molecules, and carries out a comparison of SQD energies to Full Configuration Interaction (FCI) reference energies computed in the same computational model. The reported mean absolute errors are approximately 0.0345 mHa for H_2O and 0.0138 mHa for LiH. In the tested settings these results are very close to the stated FCI references, but should be interpreted in the light of the minimal basis, the small systems, the simulation sampling and the limited number of repeated sweeps. Synthetic bit flips are treated as a controlled perturbation of measured bitstrings, not as a complete model of quantum-hardware noise.

1 Project Motivation and Aim

Molecular ground-state energy estimation is a foundational problem in computational chemistry because it provides information about molecular electronic structure. Quantum algorithms are being explored as possible tools for electronic-structure calculations, but their practical value depends not only on an algorithm's best result but also on the reliability and repeatability of its outputs.

This project uses SQD as a practical case study connecting theoretical physics with computational experimentation. The emphasis is on understanding how sampling and pipeline choices influence the configurations retained for diagonalization, how results vary across random seeds, and how synthetic bit-flip perturbations affect the final estimate.

Aim

To investigate how selected sampling and pipeline settings affect SQD estimates of the ground-state energies of H_2O and LiH.

Objectives

- Compare energy estimates across selected shot counts.
- Examine how subspace-selection settings affect the resulting calculation.
- Measure repeatability across five random seeds.
- Explore how synthetic bit-flip perturbations affect sampled configurations and energy convergence.
- Document a reproducible workflow and identify the limitations of the present experiments.

2 Methodology

2.1 SQD workflow

The workflow prepares and samples a quantum circuit whose measured bitstrings encode candidate electronic configurations. The SQD procedure processes those samples, applies electronic constraints such as particle-number and spin consistency, constructs a reduced subspace, and diagonalizes the molecular Hamiltonian in that subspace. The lowest eigenvalue obtained from the retained subspace is used as the SQD ground-state energy estimate.

Because diagonalization is restricted to the selected subspace, the result depends on which configurations are sampled and retained. Increasing the number of shots may reveal more configurations, but it does not guarantee a monotonic improvement in energy for every finite sample or random seed.

2.2 Computational environment

The experiments used simulator-based sampling. The reported setup does not establish performance on real quantum hardware. Runtime values are wall-clock measurements from Google Colab and may vary with available resources.

Table 1: Computational environment and settings.

Component	**Version / setting**
Platform	Google Colab
Qiskit	2.5.1
Qiskit Aer	0.17.2
qiskit-addon-sqd	0.12.1
FFsim	0.0.84
PySCF	2.14.0
NumPy	2.5.4
Basis set	STO-3G
Frozen orbitals	`n_frozen = 0`, as reported for the H_2O and LiH results
Ansatz repetitions	`n_reps = 2`
SQD recovery	Up to six iterations

Table 2: FCI reference energies (STO-3G).

Molecule	**FCI reference energy**
H_2O	−75.012578 Ha
LiH	−7.882324 Ha

2.3 Reference energies and error metric

The FCI values are exact diagonalization references for the stated finite STO-3G model, not experimental molecular energies. The absolute energy error is defined as

$$\Delta E_{\text{abs}} = |E_{\text{SQD}} - E_{\text{FCI}}| \,. \tag{1}$$

Since 1 Ha = 1000 mHa, energy differences are converted to millihartree (mHa) by multiplying the Hartree difference by 1000.

3 Results

3.1 Shot-count and subspace sweeps

The project explored shot counts from 5,000 to 30,000 in steps of 5000. The analysis reports that estimates stabilized relatively quickly for both molecules and that improvements beyond approximately 15,000 shots were smaller than at lower shot counts, while sampling required additional time.

The subspace sweep varied samples per batch and the number of batches. The reported interpretation is that subspace construction materially affects which configurations are available to the diagonalization step. Because the full per-setting results were not included in the supplied draft, the magnitude of each setting's effect should not be inferred from the qualitative description alone.

3.2 Repeatability across random seeds

The pipeline was run with five random seeds: 11, 22, 33, 44 and 55. The reported summary statistics are shown below.

Table 3: Five-seed summary statistics.

Metric	**H_2O**	**LiH**
Mean SQD energy	-75.012544 Ha	-7.882311 Ha
Reported standard deviation	≈ 0.000038 Ha	≈ 0.000002 Ha
Mean absolute error vs FCI	≈ 0.0345 mHa	≈ 0.0138 mHa
Mean runtime	3.26 s	2.12 s
Mean subspace dimension	275.8	94.0

The reported means lie close to the FCI reference energies, and the reported variation across seeds is small for these test cases. This is evidence of repeatability under the specific tested settings, not a guarantee that the workflow will have similar variance for larger molecules, different basis sets, or hardware execution.

3.3 Synthetic bit-flip perturbation

Synthetic bit-flip perturbations were applied to sampled bitstrings with probabilities in the range $p \in [0.00, 0.10]$. The report records that perturbed samples could increase the number and variety of observed bitstrings and, in some tested cases, move the estimated energy closer to the FCI reference. A larger candidate pool can sometimes expose configurations that were absent from the original sample, but bit flips can also create physically invalid configurations or distort sample frequencies.

Therefore, the observation should not be interpreted as evidence that noise improves SQD in general. The perturbation was synthetic and applied to sampled bitstrings; it does not represent a comprehensive hardware noise model. A robust conclusion would require repeated, paired runs across noise probabilities and seeds, together with explicit reporting of particle-number validity, retained-subspace size, energy error and runtime.

4 Discussion

The experiments illustrate the central trade-off in SQD: the quality of the final estimate depends on the quality and diversity of the sampled configurations, while the computational diagonalization is performed only in the subspace assembled from those configurations. More shots and larger subspaces can provide additional candidate states, but they also increase sampling or classical processing cost. Consequently, energy accuracy should be evaluated alongside runtime and subspace size rather than in isolation.

The reported five-seed statistics are encouraging for the two small molecular examples. However, the absolute errors are computed relative to FCI energies within the same STO-3G model. Agreement with this reference assesses the SQD procedure within that model; it does not by itself establish chemical accuracy relative to experiment or accuracy in a larger basis set.

The noise results are particularly important to interpret cautiously. A bit-flip perturbation may increase sample diversity, but it also changes the measured data artificially. If an energy estimate improves after perturbation, that outcome may reflect a more favorable finite subspace in that run rather than a general beneficial effect of noise. The present analysis is best viewed as exploratory evidence motivating a controlled follow-up experiment.

5 Limitations

- **Small-system scope:** only H_2O and LiH were considered, using the minimal STO-3G basis.
- **Simulator-only evaluation:** the sampling workflow was run in a simulated environment rather than on physical quantum hardware.
- **Limited replication:** the report records five seeds for the repeatability study, while shot-count and subspace sweeps were reported as single-seed experiments.
- **Simplified noise model:** independent synthetic bit flips do not capture the full range of hardware errors, including gate errors, readout calibration effects, decoherence and correlated noise.

6 Conclusion

This project implemented and explored an SQD workflow for estimating the ground-state energies of H_2O and LiH using the STO-3G basis and the Qiskit ecosystem. The reported five-seed results show close agreement with the corresponding FCI reference energies for the tested settings, with mean absolute errors of approximately 0.0345 mHa for H_2O and 0.0138 mHa for LiH. The experiments also suggest that shot count, subspace construction and synthetic perturbations can influence the final estimate. These findings are preliminary and are limited to the tested molecules, model and simulator settings.

7 Resources and Declarations

Software and resources

Google Colab, Qiskit, Qiskit Aer, qiskit-addon-sqd, FFsim, PySCF, NumPy, pandas, Matplotlib and openpyxl.

Declaration of AI assistance

Generative AI (ChatGPT) was used to support technical learning, debugging, explanation of SQD concepts, interpretation of selected outputs, organization of the report and editing. The team remains responsible for running and checking the code, numerical outputs, interpretations and final submission.

Team declaration

We declare that this report describes calculations and outputs generated by Team CG (Glory Ogunsola and Chiagoziem Mushinwa). Numerical claims should be read alongside the limitations stated in this report.

References

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